Renewal Radar final project report

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Purpose and outcome

Renewal Radar helps procurement and legal operations teams identify contract notice deadlines before an unwanted renewal. My intended user manages approximately 150 vendor contracts across shared folders and spreadsheets. An expiry date alone is insufficient: a contract expiring in December may require notice in October. The prototype extracts renewal obligations, calculates notice boundaries, and presents evidence and review flags in a dashboard. The results show that Renewal Radar can help users review contracts and identify deadlines, although its ability to prevent missed renewals still needs to be tested in practice.

The main change from the current spreadsheet workflow is that users can start from contract wording and see which obligations deserve attention. Search, status filters, contract cards, a monthly timeline, and a CSV export support that task. The interface separates checked date inputs from unchecked AI results. It displays saved extractions; opening the website does not run the model or send reminders. A person remains responsible for confirming the current agreement and deciding whether to act.

Design and technical choices

I used Claude Sonnet 4.6 through OpenRouter with a frozen version 6 extraction prompt. The input is page-numbered contract text. The model returns dates and term wording, renewal category, notice requirements, evidence, calculability, and reasons for human review. The saved predictions are normalized into structured inputs for a deterministic date engine. That engine subtracts calendar days or months, handles explicit deadlines and notice windows, and compares the result with a selected check date.

This division follows the professor's feedback: language interpretation belongs to the LLM, while date arithmetic belongs to code. It makes calculations reproducible and lets a reviewer inspect the source of a deadline. I rented the language model capability and built the normalization, calculation, evaluation, and interface layers. I did not implement retrieval augmented generation or an autonomous agent because the prototype supplies the document directly and does not need to execute a contractual decision.

Data and evaluation design

My initial plan was to use 138 development contracts and 40 held-out contracts from CUAD. To keep manual labelling manageable within the project timeline, I worked with 34 development contracts. The historical evaluation covers the original 40 contracts. Following the professor's advice, I examined renewal classification, expiry information, and notice requirements separately instead of assigning one combined correctness score. Binary auto-renewal scoring excludes conditional and unclear categories because forcing them into yes or no would distort the comparison.

Many original documents lacked a sufficiently confirmed date or contained several notice rights. I subsequently tested ten and then fifteen public contracts or amendments with clearer visible dates. These were purposefully selected, and many involve similar fund agreements. The final dashboard contains

those 25 additions plus 15 retained original contracts. This replacement dataset was assembled after evaluation; its results are descriptive and cannot be presented as a new blind test. The package preserves earlier predictions, reference inputs, evaluation explanations, and source files for the final 40.

The proposal set qualitative goals rather than a numerical acceptance threshold. I therefore report observed performance without inventing a pass mark afterward. Expiry evaluation on the additions checks a stated date in the supplied document, not whether it remains the operative date today. Manual reference judgments also introduce interpretation risk. The package includes archived scores for the original evaluation, but excludes the 25 original PDFs replaced in the main dataset; independently rechecking those historical source judgments would require the earlier archive.

Results and their interpretation

On the historical 40-contract evaluation, renewal category matched the reference in 38 cases, or 95%. On the 34 eligible binary cases, the model matched all 34, compared with 21 of 34 for the always-yes baseline and 30 of 34 for the retrospective keyword baseline. These comparisons show value beyond a simple guess on this sample. They do not prove general accuracy across contract types, and the keyword comparison is retrospective rather than a separately frozen prospective benchmark.

The ten clearer additions had 10 of 10 stated expiry matches, 10 of 10 renewal-category matches, and 10 of 10 manually adjudicated primary notice matches. The next fifteen had 15 of 15 stated expiry matches, 14 of 15 category matches, and 15 of 15 primary notice matches. Primary notice scoring checks the relevant amount or date, unit, actor, and purpose. It does not establish correctness of every secondary notice, explanation, or deadline calculation. For example, Applied Optoelectronics was classified as an option although renewal required a separate agreement.

Only four historical contracts qualified for the confirmed deadline comparison, and all four matched. That denominator is too small for a strong deadline-accuracy claim. Separately, 13 historical alert tests and 40 checked-input arithmetic regression cases passed. These checks validate calculation behavior for supplied inputs; they do not validate the LLM’s extraction. The selected final 40 contain 39 renewal-category matches, but this reuses previous predictions on a selected sample.

The model requested review for 27 of the original 40 contracts, two of the additional ten, and five of the additional fifteen. The final selected set has 14 raw model flags, while the checked-input dashboard has 21 review cases. Those counts differ because source review and input readiness also affect the dashboard. The archived extraction-cost estimate was approximately US$2.59 for the original 40, or US$0.065 each and US$9.70 for 150 at the same token usage and recorded rates. This excludes review labor and is not a current price quote.

Difficulties and remaining weaknesses

Review flags were initially frequent because missing dates and multiple rights were treated conservatively. Some flags reflect legitimate complexity rather than bad extraction. TCG, for example, gives the board a rolling termination right and the adviser a fixed annual termination right. The model treated this as preventing a calculable deadline, although the adviser’s deadline can be calculated separately. A useful system must distinguish actor, purpose, trigger, and date anchor instead of collapsing all termination rights into one field.

Prompt refinement and conservative normalization improved the handling of explicit dates and alternative anchors, but errors remain. Pacific Select's notice is anchored to the next term's start, so using the previous term's end creates a one-day error. Enova's output contained inconsistent calendar arithmetic despite instructions forbidding it. These examples justify calculating independently and preserving the original output for inspection. Clearer documents reduced the observed review rate, but selection bias prevents attributing that reduction entirely to better model performance.

The proposal's calibrated confidence threshold was not implemented; the system uses rule and model review flags instead. It also lacks business-day calendars, delivery buffers, automated notifications, and verification of later amendments or current renewal cycles. Some displayed dates therefore describe historical stated terms. An amendment may need its base agreement, and an unflagged AI answer can still be wrong. A review flag creates useful visibility, but its business value depends on whether the user can resolve it quickly. I did not measure that review effort or the number of unwanted renewals avoided.

Conclusion and where this goes next

Renewal Radar shows how AI extraction and deterministic date calculations can help users identify contract notice deadlines and focus their review. The results are promising, particularly for documents with clearly stated dates, but complex notice rights and uncertain terms still require human checking. The project provides a working prototype with traceable evidence and reproducible evaluations.

The next step is to track each party's notice rights separately and test the system on a fresh, independently held-out dataset. A future version could support document uploads and automated reminders once current terms and deadlines are verified. A business pilot would then measure whether it reduces review time and helps prevent missed notice windows.

Evidence files

Submitted package: evaluation/README.md; results/reproduced/evaluation_summary.json; evaluation/Additional15_Source_Review.md; docs/PRODUCT.md; datasets/final40/catalog.json. Proposal and intermediate professor feedback are retained in docs.